projectile motion

Subject: Physics

Grade: Grade 11

Here's a clean, quick summary of **projectile motion**:

* It is the motion of an object thrown into the air under **gravity only** (no air resistance).
* The motion splits into two parts:

  * **Horizontal** -> constant velocity
  * **Vertical** -> accelerated motion due to gravity
* The path followed is a **parabola**.
* Key formulas:

  * Time of flight: ($T = \frac{2v_0 \sin\theta}{g}$)
  * Maximum height: ($H = \frac{v_0^2 \sin^2\theta}{2g}$)
  * Range: ($R = \frac{v_0^2 \sin(2\theta)}{g}$)
* Maximum range occurs at **45°**, and complementary angles (like 30° and 60°) give the same range.

If you remember just one thing: ? treat x and y motions separately.

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Projectile motion is one of those topics that looks simple at first-but once you dig in, it becomes really powerful (and actually fun). Let's walk through it clearly and build your intuition step by step.

? What is Projectile Motion?

Projectile motion is the motion of an object that is thrown or launched into the air and then moves under the influence of gravity alone (ignoring air resistance).

Examples:

* A ball thrown into the air ?
* A basketball shot ?
* A stone thrown at an angle

? Key Idea (Very Important)

Projectile motion can be split into **two independent motions**:

1. **Horizontal motion (x-direction)** -> constant velocity
2. **Vertical motion (y-direction)** -> accelerated motion due to gravity

Gravity only affects the vertical direction!

? Main Equations

1. Horizontal Motion

No acceleration:

$$
x = v_0 \cos(\theta) \cdot t
$$

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2. Vertical Motion

Acceleration = (g) downward:

$$
y = v_0 \sin(\theta) \cdot t - \frac{1}{2} g t^2
$$

? Trajectory Equation (Path)

This gives the shape of the path (a parabola):

?genui?{"math_block_widget_always_prefetch_v2":{"content":"y = x\tan(\theta) - \frac{g x^2}{2 v_0^2 \cos^2(\theta)}"}}?

? This is why projectiles always follow a **parabolic path**.

? Important Quantities

1. Time of Flight (Total time in air)

[
T = \frac{2 v_0 \sin(\theta)}{g}
]

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2. Maximum Height

[
H = \frac{v_0^2 \sin^2(\theta)}{2g}
]

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3. Range (Horizontal Distance)

[
R = \frac{v_0^2 \sin(2\theta)}{g}
]

? Important Concepts

?? 1. Angle for Maximum Range

* Maximum range occurs at **45°**

?? 2. Same Range Angles

* Angles like **30° and 60°** give the same range

?? 3. At Maximum Height

* Vertical velocity = 0
* Horizontal velocity stays the same

?? Common Mistakes Students Make

* Mixing up sin and cos (very common ?)
* Forgetting that **horizontal velocity is constant**
* Using (g = 9.8 , m/s^2) with wrong sign (usually take downward as negative)

? Small Example

If a ball is thrown at angle (\theta) with speed (v_0):

* Horizontal velocity = (v_0 \cos\theta)
* Vertical velocity = (v_0 \sin\theta)

Then solve x and y separately.

? Why This Topic Matters

Projectile motion connects to:

* Engineering (rockets, missiles ?)
* Sports (shooting angles ??)
* Physics exams (very common problem type!)